

Troubleshooting: Interface and Cable Issues

1. User EXEC Mode

- **The Prompt:** Router>
- **What it does:** This is the default mode when you first log in. It has very limited capabilities. You can run basic troubleshooting commands like **ping** or **traceroute** and view a limited number of show commands, but you cannot view the device configuration or make any changes.
- **How to move to the next mode:** Type (enable)

2. Privileged EXEC Mode (Enable Mode)

- **The Prompt:** Router#
- **What it does:** This mode gives you full viewing rights to the router. You can run all show and debug commands, reboot the router (reload), and save your configuration.
- **How to move to the next mode:** Type configure terminal (often abbreviated as conf t)

3. Global Configuration Mode

- **The Prompt:** Router(config)#
- **What it does:** This is where you actually begin configuring the device. Commands entered here affect the entire router globally. For example, you would use this mode to change the device's hostname, create banners, or set administrative passwords.
- **How to move to the next mode:** Type a command targeting a specific component, such as interface gigabitEthernet 0/1 or line vty 0 4.

4. Specific Configuration Modes (Sub-modes)

From Global Configuration mode, you can dive deeper into specific components of the router.

- **Interface Configuration Mode:**

- **Prompt:** Router(config-if)#
- **Use:** Configures settings for a specific physical or virtual port (e.g., assigning IP addresses, turning the port on with no shutdown, or fixing the speed/duplex).

Line Configuration Mode:

- **Prompt:** Router(config-line)#
- **Use:** Configures how administrators connect to the device (e.g., setting up SSH/Telnet access or console port passwords).

Router Configuration Mode:

- **Prompt:** Router(config-router)#
- **Use:** Configures routing protocols like OSPF, EIGRP, or BGP.

Navigating Backwards

If you need to back out of a mode, use these commands:

- **exit:** Moves you back exactly one level (e.g., from Interface mode back to Global Config mode).
- **end (or pressing Ctrl+Z):** Drops you entirely out of any configuration mode and takes you straight back to Privileged EXEC mode (Router#).

Troubleshooting physical and Data Link layer issues

1. Speed, Duplex, and Collisions

These issues usually happen when auto-negotiation fails between two connected devices (like a router and a switch, or a PC and a switch).

- **Speed Mismatch:** Both ends of a link must operate at the same speed (e.g., 100 Mbps or 1000 Mbps). If one side is hardcoded to 1000 Mbps and the other to 100 Mbps, the link will fail, and the interface will typically go into a "down/down" state.
- **Duplex Mismatch:** Happens when one side of a connection is set to Full-Duplex and the other is set to Half-Duplex. The link will stay up, but performance will be terrible.
- **Collisions:** In a Half-Duplex environment (or when there is a duplex mismatch), collisions occur when two devices try to send data at the exact same time. Late Collisions almost always point to a duplex mismatch or a cable that exceeds the maximum distance length.

show interfaces GigabitEthernet0/1

show interfaces status

#For Speed/Duplex Mismatches:

Look for: Full-duplex, 1000Mb/s, link type is auto, media type is 10/100/1000BaseTX

For Collisions:

Look near the bottom of the output for the error counters: 0 runs, 0 giants, 0 throttles 0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored 0 output errors, 0 collisions, 0 interface resets 0 late collisions, 0 deferred

Note: If you see the late collisions counter climbing, you almost certainly have a duplex mismatch where one side is Full-Duplex and the other is Half-Duplex.

```

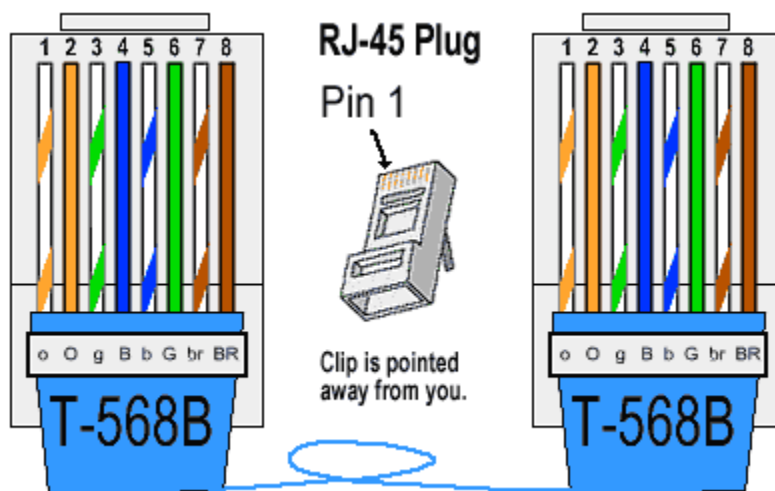
CiscoSwitch# show interfaces GigabitEthernet0/1
GigabitEthernet0/1 is up, line protocol is up (connected)
Description: Uplink to Core Switch (Gi1/0/1)
Hardware is Gigabit Ethernet, address is 00fe.b2c3.d4e5 (bia
00fe.b2c3.d4e5)
MTU 1500 bytes, BW 1000000 Kbit, DLY 10 usec, reliability 255/255,
txload 1/255, rxload 1/255
Full-duplex, 1000Mb/s, link type is auto, media type is
10/100/1000BaseTX
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
5 minute input rate 1042000 bits/sec, 981 packets/sec
5 minute output rate 1130000 bits/sec, 1075 packets/sec
21432 packets input, 19876543 bytes, 0 no buffer
Received 150 broadcasts (92 multicasts)
0 runs, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
23541 packets output, 21987654 bytes, 0 underruns
0 output errors, 0 collisions, 0 interface resets
0 babbles, 0 late collisions, 0 deferred
0 lost carrier, 0 no carrier, 0 pause output

CiscoSwitch#

```

2. Copper Cable Issues

Copper cables (Unshielded Twisted Pair, or UTP) are the standard for connecting end devices to switches.



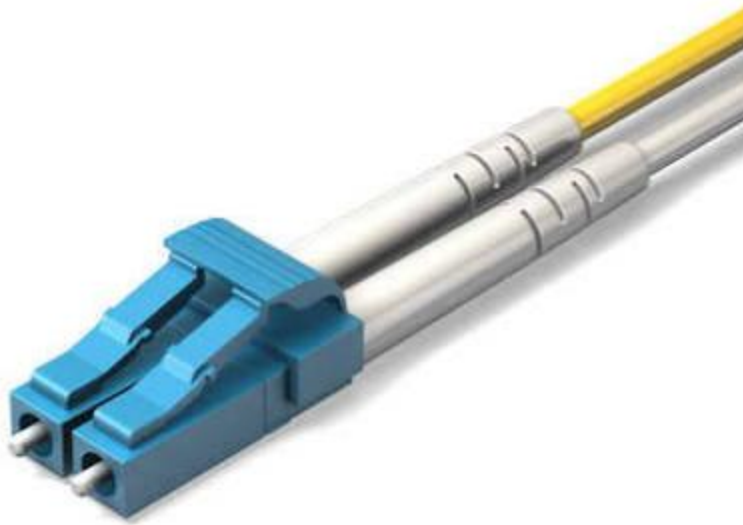
- **Pin-out Issues (Straight-through vs. Crossover):**
 - Straight-through cables: Used to connect unlike devices (e.g., a PC to a switch). Transmit pins map to receive pins.
 - Crossover cables: Used to connect like devices (e.g., switch to switch). Physically crosses transmit and receive wires.

- Note: Modern devices use Auto-MDIX to automatically detect and correct cable pinouts.
- **Cable Types:**
 - Cat5e: Supports up to 1 Gbps.
 - Cat6/Cat6a: Supports up to 10 Gbps.
- **Distance limitations:** Standard copper Ethernet has a strict maximum distance of 100 meters (328 feet). Going beyond this causes signal degradation and late collisions.

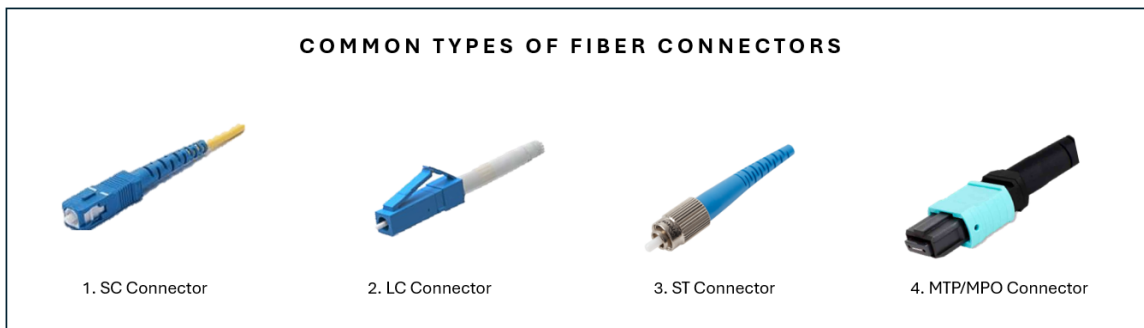
3. Fiber Optic Issues

Fiber uses light instead of electrical signals, making it immune to electromagnetic interference (EMI) and capable of spanning much longer distances.

- **Cable Types (SMF vs. MMF):**
 - Single-Mode Fiber (SMF): Uses a laser, has a very small core, and is used for long distances. Usually features a yellow jacket.
 - Multi-Mode Fiber (MMF): Uses an LED, has a thicker core, and is used for shorter distances. Usually features an aqua or orange jacket.
- **Signal Levels (Attenuation/Loss):** Fiber connections require precise light levels. If a cable is bent too sharply, is dirty, or the run is too long, the signal weakens (dBm). If the receive signal is too weak, the link will fail.
- **Connectors:**
 - LC (Lucent Connector): Small form-factor, very common in modern switches.



- SC (Subscriber Connector): Older, square, push-pull connector.



- ST (Straight Tip): Uses a bayonet-style twist-and-lock mechanism.

4. Interface States and Errors

When you use the 'show interfaces' command on a Cisco router or switch, you need to interpret the output.

Error/State	What it usually means
Administratively Down	The administrator literally typed `shutdown` on the interface.
Down / Down	A Layer 1 physical issue. The cable is unplugged, cut, or plugged into a device that is powered off.
Up / Down	A Layer 2 issue. The physical cable is fine, but there is a framing, encapsulation, or clock-rate mismatch.
CRC Errors	Cyclic Redundancy Check errors mean frames are arriving corrupted. Usually caused by a bad cable or electrical interference (EMI).
Runts / Giants	Runts are frames smaller than the 64-byte minimum (usually caused by collisions). Giants are frames larger than the 1518-byte maximum (usually a bad NIC or MTU mismatch).

Cloud Computing

1.Hypervisor

A hypervisor is software (or firmware) that allows one physical computer/server to run multiple virtual machines (VMs).

Main role

- Shares physical resources such as CPU, RAM, storage, and network interfaces.
- Creates and manages virtual machines.
- Keeps VMs isolated from each other.
- Allocates resources to each VM.
- Allows different operating systems to run on the same physical server.

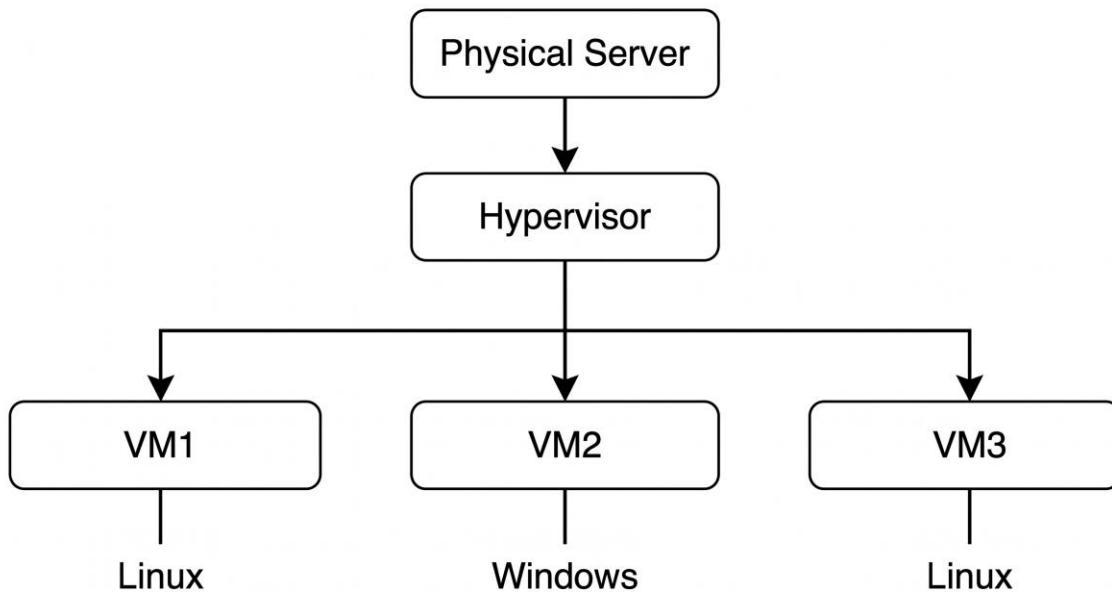
Types of hypervisors

Type	Description	Example
Type 1 – Bare Metal	Runs directly on physical hardware	VMware ESXi, Microsoft Hyper-V, KVM
Type 2 – Hosted	Type 2 – Hosted	VirtualBox, VMware Workstation

2. Virtual Machine (VM)

A virtual machine is a software-based computer running on a physical computer through a hypervisor.

For example, one physical server could run:

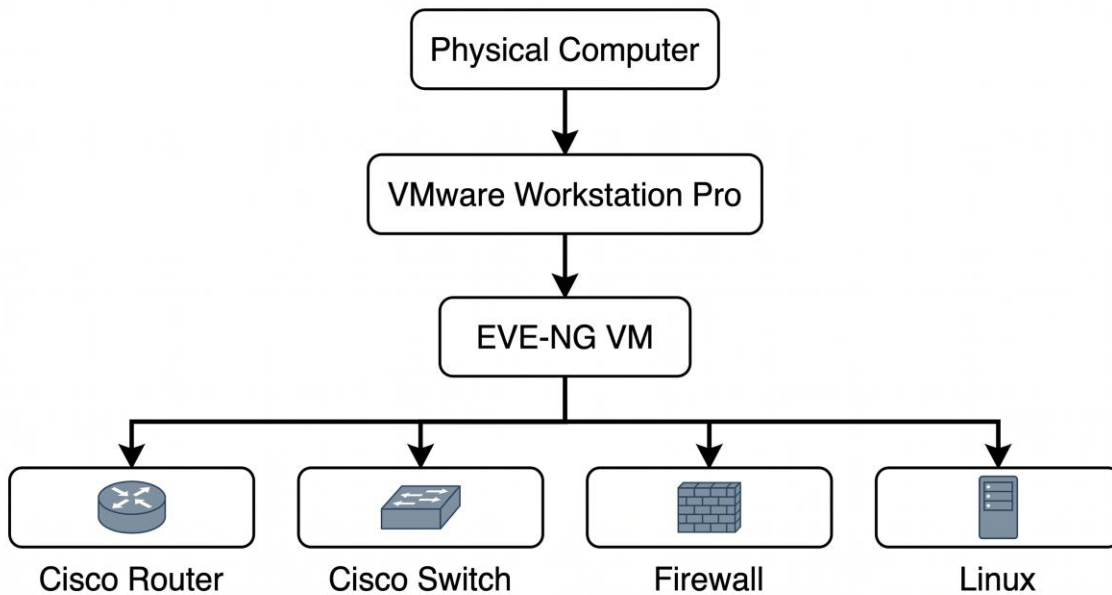


Each VM can have its own:

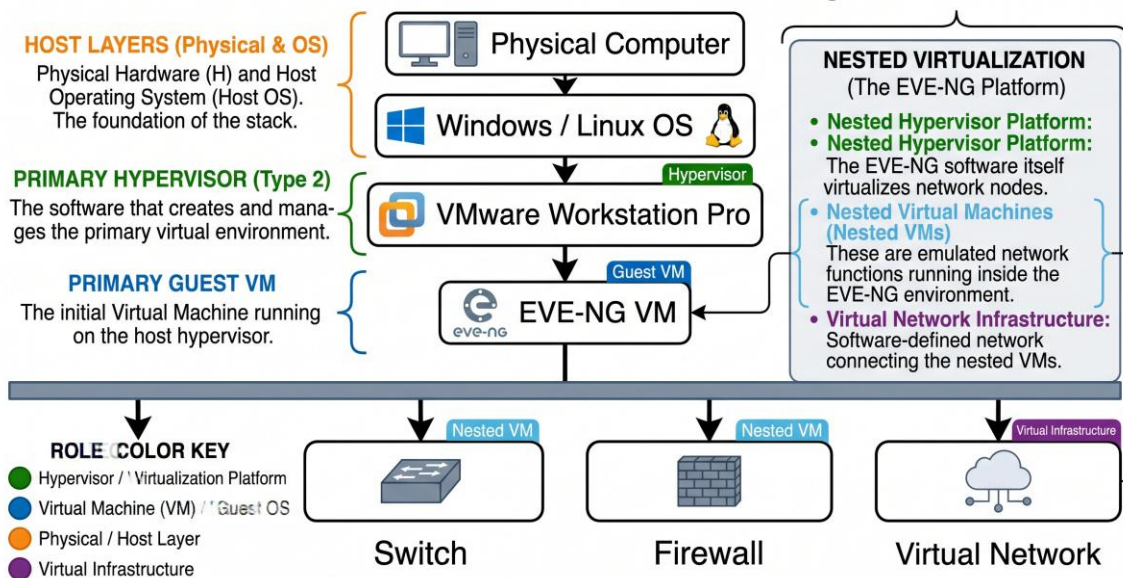
- Operating system
- Virtual CPU
- Virtual RAM
- Virtual disk
- Virtual network interface

- Applications

Example of Hypervisor and VMware



Virtualization Stack Role Identification Diagram

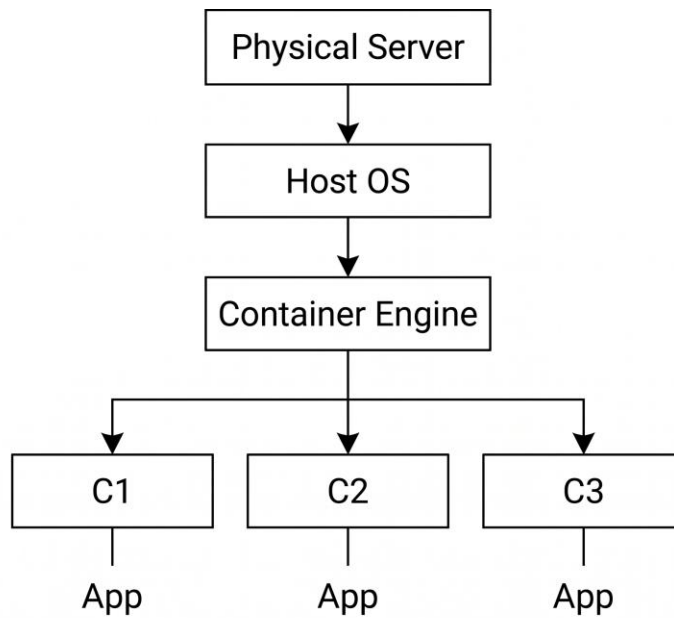


3. Containers

A container is a lightweight, isolated environment used to run an application and its required dependencies.

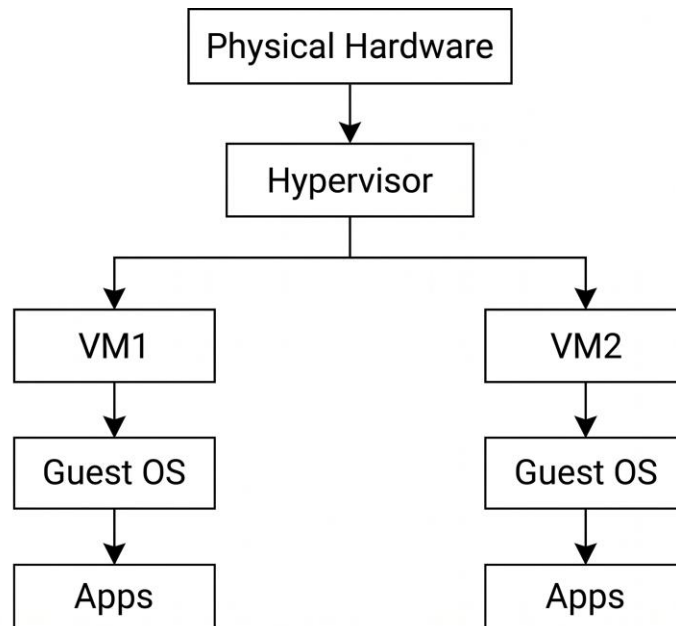
Unlike a VM, a container does not normally contain a complete guest operating system.

Containers share the **host operating system kernel**, while keeping applications isolated.



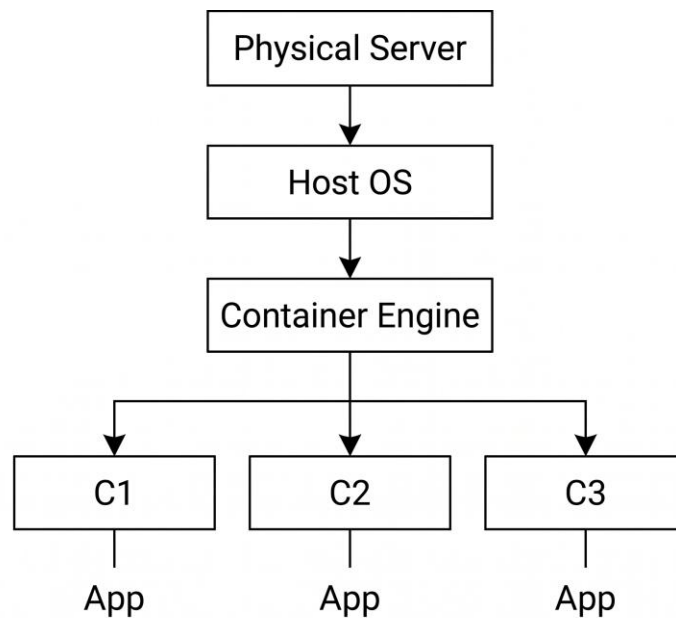
Container vs Virtual Machine

Virtual Machine



Every VM normally has its own complete guest operating system.

Container



Containers share the host OS kernel.

That's why containers are generally much smaller and faster to start than VMs.

Docker Engine

Docker Engine is the core technology that runs and manages containers.

It handles things such as:

- Creating containers
- Starting/stopping containers
- Networking
- Storage
- Resource management

Kubernetes

Kubernetes (K8s) is an open-source container orchestration platform used to deploy, manage, scale, and monitor containers across multiple servers. The easiest way to remember:

Docker runs containers. Kubernetes manages many containers.

Wireless Principles

1.5.a Band and Channel Selection

What is a wireless band?

A frequency band is a range of radio frequencies used by Wi-Fi.

The main Wi-Fi bands are:

Band	Common Wi-Fi	Characteristics
2.4 GHz	802.11b/g/n/ax	Longer range, better wall penetration, more interference
5 GHz	802.11a/n/ac/ax	Faster, more channels, less interference, shorter range
6 GHz	Wi-Fi 6E/7	Many clean channels, high speed, shorter range

What is a channel?

A channel is a smaller portion of a frequency band used by a wireless network.

Example:

Band = Highway

Channel = Lane on the highway

2.4 GHz channels

The 2.4-GHz band has overlapping channels. In many regulatory domains, the commonly used non-overlapping set is:

Channel 1 → Channel 6 → Channel 11

Example:

2.4 GHz

↓

[Ch 1] [Ch 6] [Ch 11]

↑

↑

↑

Non-overlapping channels

Using overlapping channels can cause co-channel/adjacent-channel interference.

5 GHz

5 GHz provides more channels and generally has less congestion than 2.4 GHz.

1.5.b RF Characteristics

RF = Radio Frequency

Wi-Fi uses radio waves to transmit data through the air.

Important RF characteristics:

1. Frequency

Frequency tells us how many times a radio wave cycles per second.

Measured in:

Hz → kHz → MHz → GHz

For example:

2.4 GHz = 2.4 billion cycles per second

2. Wavelength

Frequency and wavelength are inversely related:

Higher frequency → shorter wavelength

Lower frequency → longer wavelength

Therefore:

2.4 GHz → longer wavelength → generally better penetration/range

5 GHz → shorter wavelength → generally shorter range

3. Amplitude / Power

Amplitude represents the strength of the radio signal.

Higher transmitted power generally means a stronger signal, but increasing power does not automatically solve Wi-Fi problems because interference and regulatory limits still matter.

4. Signal Strength

Signal strength is commonly measured in dBm.

Example:

-30 dBm → Very strong

-50 dBm → Strong

-67 dBm → Good for many applications

-80 dBm → Weak

-90 dBm → Very weak

Important: dBm values are negative. A value closer to zero is stronger.

So:

-40 dBm is stronger than -70 dBm.

5. Attenuation(reduction) अटेन्युएसन

Attenuation = loss of signal strength

Signal becomes weaker because of:

- Distance
- Walls
- Floors
- Furniture
- Metal objects
- Glass
- Other obstacles

Example:

AP —————→ Wall —————→ Laptop

Strong signal

Weaker signal

1.5.c Security Protocols

Wireless networks need security because communication travels through the air.

WEP

WEP = Wired Equivalent Privacy

- Old wireless security protocol
- Uses weak encryption

- Easily compromised
- Obsolete

Do not use WEP on modern networks.

WPA

WPA = Wi-Fi Protected Access

Developed as an improvement over WEP.

However, WPA is now considered outdated.

WPA2

WPA2 uses AES/CCMP and became widely used for secure Wi-Fi.

Two common modes:

WPA2-Personal

- Uses a pre-shared password
- Common for home/small networks

WPA2-Enterprise

- Uses authentication infrastructure such as 802.1X/RADIUS
- Common in organizations

WPA3

WPA3 is the newer Wi-Fi security standard.

Two common modes:

WPA3-Personal

- Uses SAE instead of the older WPA2-PSK authentication method

WPA3-Enterprise

- Designed for enterprise networks
- Uses stronger authentication/security mechanisms

Easy security evolution

WEP

↓

WPA

↓

WPA2

↓

WPA3

Remember: For modern deployments, prefer WPA2 or WPA3, depending on device compatibility and security requirements.

1.5.d Causes of Interference

Interference = unwanted RF energy that disrupts wireless communication.

There are two major categories you should understand

1. Co-channel interference

Two or more APs use the same channel and compete for airtime.

Example:

AP1 — Channel 6

↕

AP2 — Channel 6

They are on the same channel.

2. Adjacent-channel interference

Nearby channels overlap and interfere with each other.

Example:

AP1 → Channel 1

AP2 → Channel 2

Depending on the channel arrangement and regulatory domain, overlapping channels can create interference.

Common sources of Wi-Fi interference

Other Wi-Fi networks

Neighboring wireless networks can use the same or overlapping channels.

Bluetooth devices

Bluetooth operates in the 2.4-GHz band and can contribute to interference.

Microwave ovens

Microwave ovens operate around 2.4 GHz and can interfere with 2.4-GHz Wi-Fi.

Cordless phones

Some older cordless phones operate in the 2.4-GHz range.

Physical obstacles

Walls and metal objects don't necessarily create RF "interference," but they can cause attenuation and multipath effects, reducing signal quality.

1.6. Troubleshoot wired and wireless client connectivity (IP configuration, network

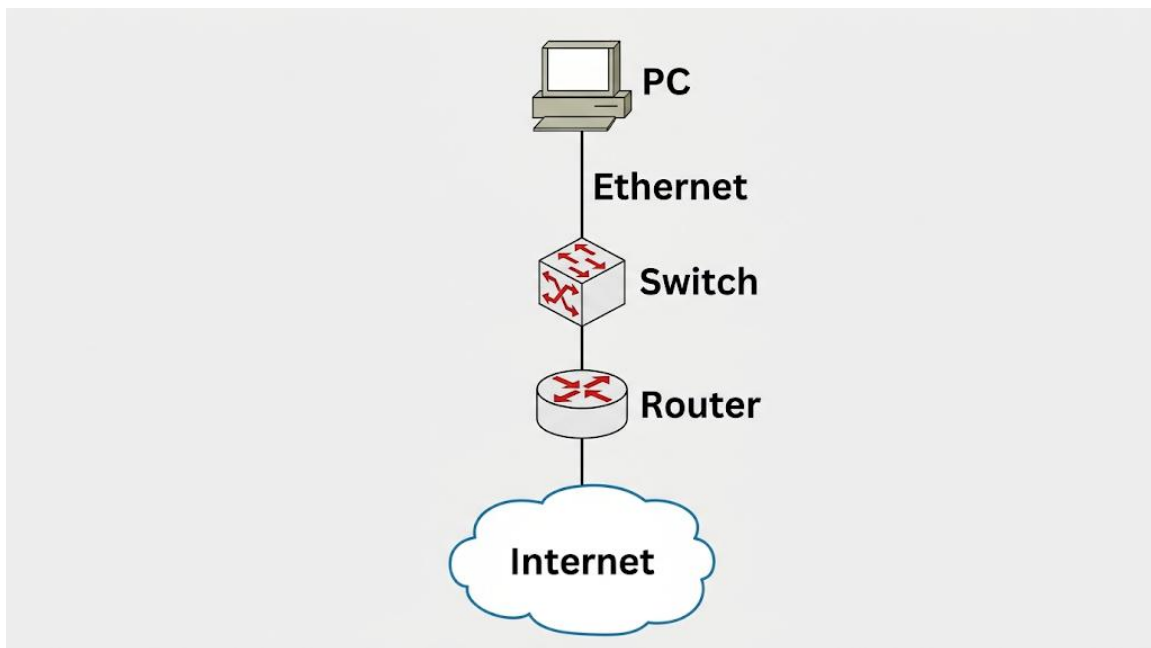
reachability, and wireless security parameters on Windows, MacOS, and Linux)

1. Basic Troubleshooting Method

Common problems

- Ethernet cable disconnected
- Bad cable
- Switch port disabled
- Incorrect VLAN
- NIC disabled
- Wrong IP address
- Wrong subnet mask
- Wrong default gateway
- DHCP failure
- DNS problem
- Port-security violation

Wired Client Connectivity



Windows

Ipconfig

`ipconfig /all`

DHCP Troubleshooting

Normally a client obtains an address automatically.

ipconfig /release

ipconfig /renew

If Windows receives an address like:

169.254.x.x

it usually means the client did not successfully obtain an IPv4 address from DHCP.

This is called an APIPA(**Automatic Private ip Addressing**) address.

Possible causes:

- DHCP server unavailable
- DHCP relay problem
- VLAN problem
- Switch port problem
- DHCP scope exhausted
- Cable/Wi-Fi connectivity problem

Network Reachability

Ping

Ping Yourself

ping 127.0.0.1

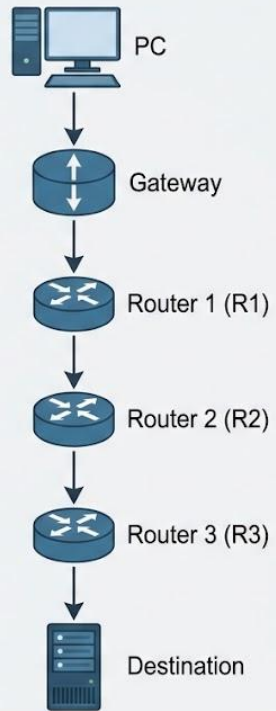
Test DNS

nslookup google.com

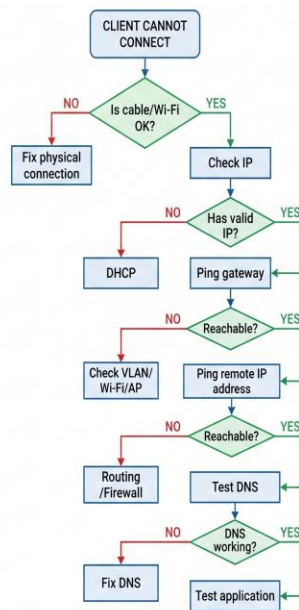
Traceroute

tracert 8.8.8.8

Network Trace Route



Troubleshooting Decision Tree



Wireless Client Troubleshooting

netsh wlan show interfaces

netsh wlan show profiles

getmac

arp -a

tracert -d 8.8.8.8

pathping